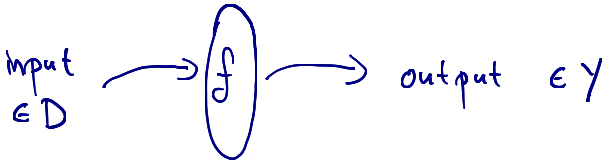
Chapter 7 Transcendental Functions

7.1 Inverse Functions and Their Derivatives

Function

A ***function*** from a set D to a set Y is a rule that assigns a unique element in Y to each element in D.



Polynomial

A polynomial is a function of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0,$$

where $n$is a nonnegative integer, $n = 0, 1, 2, 3, \ldots$ $a_i \in \mathbb{R}$

Example:

$f(x) = 3x + 1$

$f(x) = 4x^2 + 3x + 2$

$f(x) = 7x^4 + 3$

There are two types of functions.

Algebraic Functions

Constructed from polynomials using algebraic operations $+, -, \times, \div$ and taking power or roots.

Example: $f(x) = \frac{x+1}{2x+1}$

$g(x) = (4x^2+5)^5$

$h(x) = \sqrt{6x+7}$

Transcendental Functions

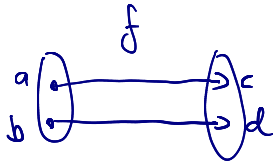
Functions that are not algebraic functions.

Example:

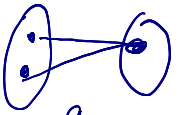
- ➢ Natural logarithmic functions $\ln(x^2+7)$
- ➢ Trigonometric functions $\sin(5x+3)$
- ➢ Hyperbolic functions
- ➢ And their inverses → $\cosh\left(\frac{x^2+x^{-2}}{2}\right)$

Definition **One-to-One Function**

A function $f(x)$ is **one-to-one (1-1)** on a domain D if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ in D.



$f(x) = x$



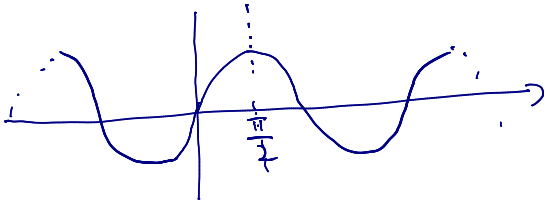
g is not 1-1

$f(x) = c$ for all $x \in D$

$f(x) = x^2$

Example:

1. $f(x) = \sqrt{x}$ is one-to-one.
   $(f : \mathbb{R}_{\geq 0} \to \mathbb{R})$
2. $f(x) = \sin x$ is *not* one-to-one on the interval $[0, \pi]$. The sine function *is* one-to-one on the interval $[0, \pi/2]$.



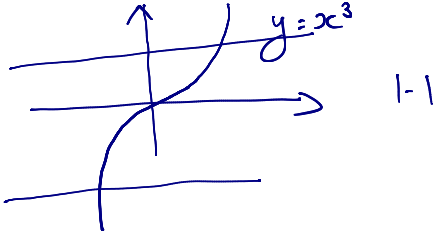
since $\sin\left(\frac{\pi}{6}\right) = \sin\left(\frac{5\pi}{6}\right)$, but $\frac{\pi}{6} \neq \frac{5\pi}{6}$.

$\sin x$ is 1-1 on $[0, \frac{\pi}{2}]$ as $\sin x$ is strictly increasing on $[0, \frac{\pi}{2}]$.

To show f is 1-1

$f(x)$ is 1-1 $\Leftrightarrow$ the graph $y = f(x)$ intersects every horizontal line ≤ 1 time.

Method 1: Use the Horizontal line test.

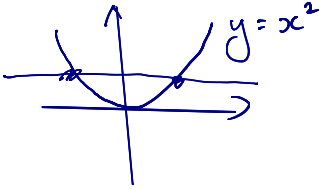


Method 2: Use the definition of an 1-1 function.

Show that $\forall\, x_1, x_2 \in D,\ x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.

Or, equivalently, show that

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$



How to show f is not 1-1?

Find **two distinct values** $x_1, x_2 \in D$ such that

$$f(x_1) = f(x_2) \text{ but } x_1 \neq x_2.$$

Example:

Is $f(x) = 1 + \sin x$ an 1-1 function? No.

$f(0) = f(\pi) = 1$ but $0 \neq \pi$.

Example:

Show that $f(x) = x^5 + 2$ is 1-1.

Suppose $f(x_1) = f(x_2)$. We want to show that $x_1 = x_2$.

$f(x_1) = f(x_2) \Rightarrow x_1^5 + 2 = x_2^5 + 2$

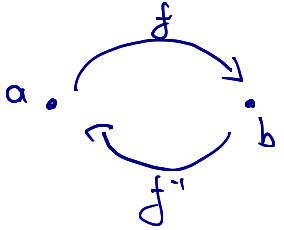
$\Rightarrow x_1^5 = x_2^5$

$\Rightarrow (x_1^5)^{1/5} = (x_2^5)^{1/5}$

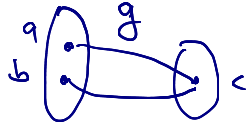
$\Rightarrow x_1 = x_2$.

$\therefore f(x)$ is 1-1.

Informally, the inverse of a function is a function that sends an output back to its input.



Only 1-1 functions can have inverses.



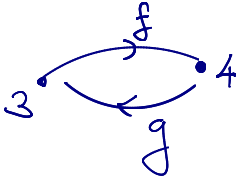
$g^{-1}(c)$

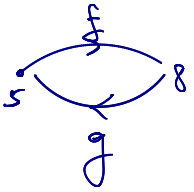
g is not 1-1 $\Rightarrow$ g has no inverse

eg. $f: \mathbb{R} \to \mathbb{R}$ & $g: \mathbb{R} \to \mathbb{R}$

$f(x) = 2x - 2$ $\quad$ $g(x) = \frac{1}{2}x + 1$

f & g are inverses of each other.





Definition **Inverse Function**

Suppose that f is a 1-1 function on a domain *D* with range *R*. The *inverse function* f^{-1} is defined by

$$f^{-1}(b) = a \text{ if } f(a) = b.$$

The domain of f^{-1} is *R* and the range of f^{-1} is *D*.

$R_{f^{-1}} = D_f$

$D_{f^{-1}} = R_f$.

Remark on Symbol

$f^{-1} \neq \frac{1}{f}$

eg. $\sin^{-1}(x) \neq \frac{1}{\sin x} = (\sin x)^{-1}$

$\sin^3(x) = (\sin x)^3$

Composing a function and its inverse

$$(f^{-1} \circ f)(x) = x \quad , \quad x \in D_f$$

$$(f \circ f^{-1})(y) = y \quad , \quad y \in D_{f^{-1}} = R_f$$

Example:

Show that the inverse of $f(x) = 5x$ is $g(x) = \frac{1}{5}x$.

To show $(f \circ g)(x) = x$ & $(g \circ f)(x) = x$

$$(f \circ g)(x) = f(g(x)) = f\left(\tfrac{1}{5}x\right) = 5\left(\tfrac{1}{5}x\right) = x$$

$$(g \circ f)(x) = g(f(x)) = g(5x) = \tfrac{1}{5}(5x) = x \; .$$

<u>Remark:</u> Only a 1-1 function can have an inverse.

Example:

Which of the following functions have an inverse?

1. $f(x) = \sqrt{x+1}$

2. $g(x) = x^2 - x$ → Not 1-1, as $g(0) = g(1) = 0$ ⇒ no inverse

3. $h(x) = \sin x,\ x \in (0, \pi)$ → Not 1-1. ⇒ has no inverse

4. $k(x) = \cos x,\ x \in [0, \pi]$ → 1-1 ⇒ so $k(x)$ has an inverse.

① $f(x) = \sqrt{x+1}$ is 1-1

($D_f = [-1, \infty)$)

Proof: $f(x_1) = f(x_2) \Rightarrow \sqrt{x_1+1} = \sqrt{x_2+1}$

$\Rightarrow x_1 + 1 = x_2 + 1$

$\Rightarrow x_1 = x_2$.

$f(x)$ is 1-1 ⇒ $f(x)$ has an inverse.

Finding Inverses

① Solve the equation $y = f(x)$

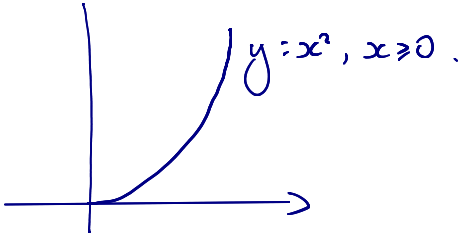
That is, write x as a function of y

$\Rightarrow x = f^{-1}(y)$

② Interchange x & y to get $y = f^{-1}(x)$.

Example:

Find the inverse of $y = f(x) = x^2, x \geq 0$.



① $y = x^2 \Rightarrow \sqrt{y} = \sqrt{x^2}$

$= |x|$

$= x$ (as $x \geq 0$)

(interchange x & y)

② $\Rightarrow y = f^{-1}(x) = \sqrt{x}$.

Question: If $y = x^2, x \leq 0$, what is the inverse of y?

$y = x^2 \Rightarrow \sqrt{y} = \sqrt{x^2} = |x| = -x \quad (\because x \leq 0)$ → $x = -\sqrt{y}$

$\Rightarrow f^{-1}(x) = -\sqrt{x}$

Example:

Find the inverse of $f(x) = \sqrt{4x - 5}$, if it exists. What is the domain of f^{-1}?

(Exercise: Check that $f(x)$ is 1-1.)

① Let $y = \sqrt{4x-5}$

$\Rightarrow y^2 = 4x - 5$

$\Rightarrow 4x = y^2 + 5$

$\Rightarrow x = \frac{y^2+5}{4}$

② $y = f^{-1}(x) = \frac{x^2+5}{4}$.

Domain of f^{-1} = $R_f = [0, \infty)$

Range of f^{-1} = $D_f = [\frac{5}{4}, \infty)$

Derivative of Inverses

Example: $f(x) = \frac{1}{2}x + 1$ $\quad f^{-1}(x) = 2x - 2$

Notice that the derivative of $f(x) = \frac{1}{2}$ &

the derivative of $f^{-1}(x) = 2$.

Observe that they are reciprocals of each other.

i.e., $\frac{d}{dx} f^{-1}(x) = \frac{1}{\frac{d}{dx} f(x)}$ (*) (for this particular example.)

It turns out that this is not a special case,

but (*) is true in general.

Derivative of Inverses

If the slope of $y = f(x)$ at (a, b) is $f'(a)$ and $f'(a) \neq 0$, then the slope of $y = f^{-1}(x)$ at (b, a) is $\dfrac{1}{f'(a)}$.

$\Rightarrow f(a) = b$

$f^{-1}(b) = a$

$$\frac{d}{dx} f^{-1}\Big|_{x=b} = \frac{1}{\frac{d}{dx} f\big|_{x=a}}$$

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.$$

Remark:

If f has a horizontal tangent line at (a, b), then f^{-1} has a vertical tangent line at (b, a) with infinite slope, implying f^{-1} is not differentiable at b.

Theorem The Derivative Rule for Inverses ~~(Page 374)~~

If f has an interval I as domain and $f'(x)$ exists and is never zero on I, then f^{-1} is differentiable at every point in its domain (the range of f). The value of $(f^{-1})'$ at a point b in the domain of f^{-1} is the reciprocal of the value of f' at the point $a = f^{-1}(b)$:

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$$

or

$$\frac{df^{-1}}{dx}\Big|_{x=b} = \frac{1}{\frac{df}{dx}\Big|_{x=f^{-1}(b)}}$$

Example: $f(x) = x^2,\ x \geq 0.$

$$f^{-1}(x) = \sqrt{x} \quad \Rightarrow \quad f'(x) = 2x$$

$$\boxed{(f^{-1})'(x) = \frac{1}{2\sqrt{x}}}$$

From the theorem,

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

$$= \frac{1}{2f^{-1}(x)} \qquad (\because f'(x) = 2x)$$

$$= \frac{1}{2\sqrt{x}}.$$

Example: $f(x) = x^3 - 2$ $\Rightarrow$ $f'(x) = 3x^2$

Find $\frac{df^{-1}}{dx}\Big|_{x=6}$ without finding f^{-1}.

$$(f^{-1})'(6) = \frac{1}{f'(f^{-1}(6))}$$

If $x = f^{-1}(6) \Rightarrow f(x) = 6$

$\Rightarrow x^3 - 2 = 6$

$\Rightarrow x^3 = 8$

$\Rightarrow x = 2$

So $(f^{-1})'(6) = \frac{1}{f'(2)} = \frac{1}{3(2)^2} = \underline{\underline{\frac{1}{12}}}$

$$\frac{d}{dx} f^{-1}\Big|_{x=6} = \frac{1}{\frac{df}{dx}\Big|_{x=f^{-1}(6)=2}} = \underline{\underline{\frac{1}{12}}}.$$

Example:

Let $f(x) = \frac{x^3}{4} + x - 1$. $\Rightarrow f'(x) = \frac{3x^2}{4} + 1 \quad -(*)$

- Find $f^{-1}(3)$.
- Find $(f^{-1})'(3)$.

• let $x = f^{-1}(3) \Rightarrow f(x) = 3$

$\Rightarrow \frac{x^3}{4} + x - 1 = 3$

$\Rightarrow x^3 + 4x - 16 = 0$

$\Rightarrow (x-2)(x^2+2x+8) = 0$

$\Rightarrow x = 2$ OR $x^2+2x+8 = 0$

$(x+1)^2 + 7 = 0$

$(x+1)^2 + 7 > 0$ for all x

$\Rightarrow x^2+2x+8 \neq 0$

$\Rightarrow \boxed{f^{-1}(3) = 2}$

$\Rightarrow (f^{-1})'(3) = \frac{1}{f'(f^{-1}(3))}$

$= \frac{1}{f'(2)}$

from $(*) \Rightarrow = \frac{1}{\frac{3(2)^2}{4} + 1} = \frac{1}{4}$